Tuning report: tropical / MaxSubsetSum

2026-10-22T19:16:57

1 Decision

Approving freezes `fishr.lambda` = 100 into every cell of the grid (frozen hash `a7b38650bd58`). The experiment YAML has `fishr.lambda` = 1000 for this campaign. `frozen.yaml` is currently unapproved with {'fishr.lambda': 100.0}.

Selection rule. The candidate with the lowest mean `id_val_loss` wins, averaged over 6 cell(s) and 2 tuning seed(s) each (global seed 1000, disjoint from the grid's 0), across 4 candidates. A candidate with a missing or diverged run is not eligible.

Verdict: REVIEW

- **review**: the candidate means differ by less than 0.1%: the metric barely responds to the knob
- **review**: the winner is not distinguishable from fishr.lambda=10: the paired gap's 95% CI is [-1.74e-08, 3.55e-06]
- **review**: fishr.lambda=10: fishr.lambda * fishr_penalty_final is 4.8e-05 of id_val_loss (median over runs); the term barely enters the loss
- **review**: fishr.lambda=100: fishr.lambda * fishr_penalty_final is 0.00048 of id_val_loss (median over runs); the term barely enters the loss
- **review**: fishr.lambda=1000: fishr.lambda * fishr_penalty_final is 0.0048 of id_val_loss (median over runs); the term barely enters the loss
- **review**: 12 of 12 winner runs reached their best id_val_loss at the last logged epoch: the search budget may be too short to rank candidates
- **review**: 12 of 12 winner runs picked their last epoch (best_epoch = 19): still improving when the search budget ran out, so candidates were ranked before converging
- **info**: the search ran with optim.epochs=20; the grid runs with optim.epochs=100
- **info**: devices: cuda:0

block: do not approve; fix the search and re-run it. **review**: read the sections below before approving. **info**: context only.

2 Search setup

- `fishr.lambda` ∈ {10, 100, 1000, 1e+04}
- Cells trained (`tune.only` {'objective': 'fishr'}):
  - `model_type=vanilla head=seq2seq objective=fishr`
  - `model_type=vanilla head=pointer objective=fishr`
  - `model_type=adaptive head=seq2seq objective=fishr`

- `model_type=adaptive head=pointer objective=fishr`
- `model_type=tropical head=seq2seq objective=fishr`
- `model_type=tropical head=pointer objective=fishr`

- Search-only overrides (never frozen): `{'optim': {'epochs': 20}}`. The chosen value was ranked under this budget, not the grid's.

3 Candidates

Table 1 ranks the candidates on `id_val_loss` (lower is better); Figure 1 plots them. Candidates share seeds, so Table 2 compares each with the winner run by run (same cell, same seed).

Table 1: Every candidate: rank, runs stored/expected, diverged, mean with 95% bootstrap CI, SD and range over all its runs.

#	`fishr.lambda`	runs	div.	mean	95% CI	SD	range	
2	10	12/12	0	0.03831	[0.012, 0.0727]	0.0589	$[1.77 \times 10^{-4}, 0.181]$	
1	100	12/12	0	0.0383	[0.012, 0.0737]	0.0589	$[1.77 \times 10^{-4}, 0.181]$	**selected**
3	1000	12/12	0	0.03831	[0.00922, 0.0737]	0.0589	$[1.77 \times 10^{-4}, 0.181]$	
4	1e+04	12/12	0	0.03831	[0.0114, 0.0736]	0.059	$[1.77 \times 10^{-4}, 0.181]$	

Table 2: Each candidate minus the winner on the runs they share, signed so a positive gap is worse than the winner. Worse/better/equal count the runs; p is a two-sided sign test.

candidate	n	gap	95% CI	worse	better	equal	p
`fishr.lambda=10`	12	1.392×10^{-6}	$[-1.74 \times 10^{-8}, 3.55 \times 10^{-6}]$	2	10	0	0.0386
`fishr.lambda=1000`	12	1.909×10^{-6}	$[1.33 \times 10^{-7}, 4.76 \times 10^{-6}]$	12	0	0	0.0005
`fishr.lambda=10000`	12	1.001×10^{-5}	$[3.32 \times 10^{-8}, 2.77 \times 10^{-5}]$	11	1	0	0.0063

4 Per cell

One value is frozen for every cell, chosen by the mean over cells. Figure 2 shows whether the winner is best in each cell or is carried by a few of them.

Table 3: The best candidate in each cell.

cell	best
`model_type=adaptive head=pointer`	fishr.lambda=10
`model_type=adaptive head=seq2seq`	fishr.lambda=10
`model_type=tropical head=pointer`	fishr.lambda=100
`model_type=tropical head=seq2seq`	fishr.lambda=10
`model_type=vanilla head=pointer`	fishr.lambda=10
`model_type=vanilla head=seq2seq`	fishr.lambda=10

5 Other training-distribution metrics

Not used for selection; shown so a winner that trades them away is visible.

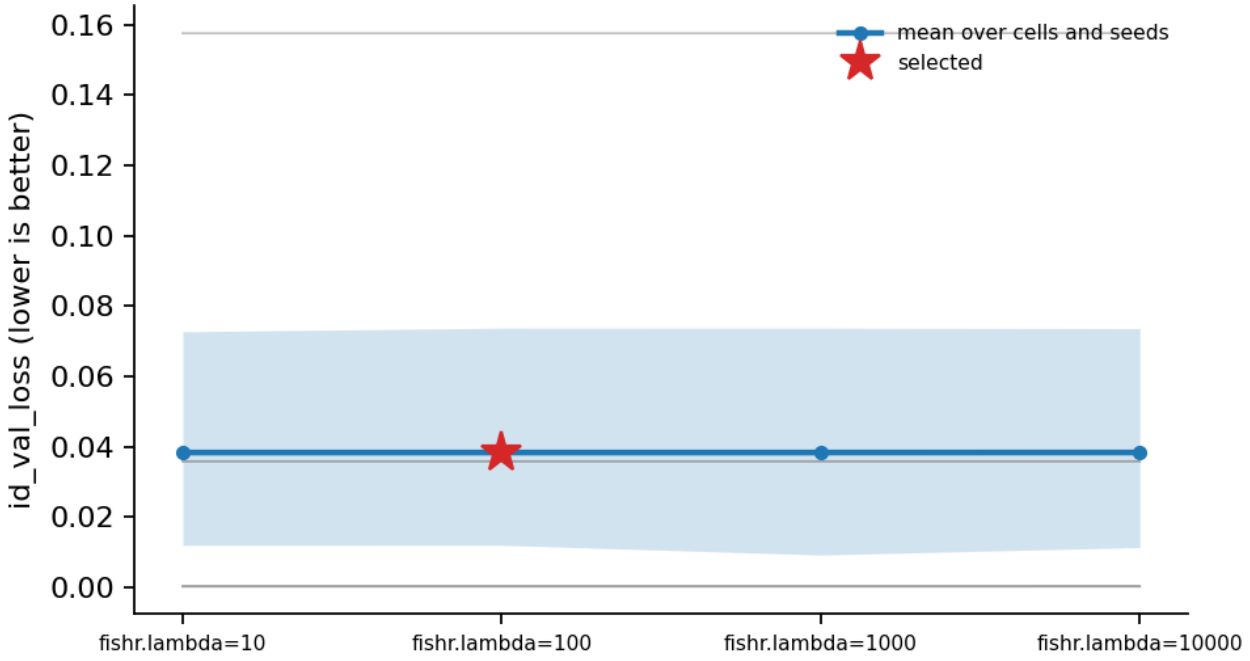


Figure 1: `id_val_loss` per candidate: the mean over every run with a 95% bootstrap CI (blue), each cell's seed-mean (grey), the selected candidate (red star).

Table 4: Mean of every other in-distribution metric per candidate.

candidate	`id_f1`	`id_exact`	`fishr_penalty_final`	`best_epoch`
fishr.lambda=10	0.993	0.992	9.79×10^{-7}	19
fishr.lambda=100	0.993	0.992	9.79×10^{-7}	19
fishr.lambda=1000	0.993	0.992	9.80×10^{-7}	19
fishr.lambda=10000	0.993	0.992	9.80×10^{-7}	19

6 Training curves

The mean over every cell and seed of each candidate; the selected candidate in bold red.

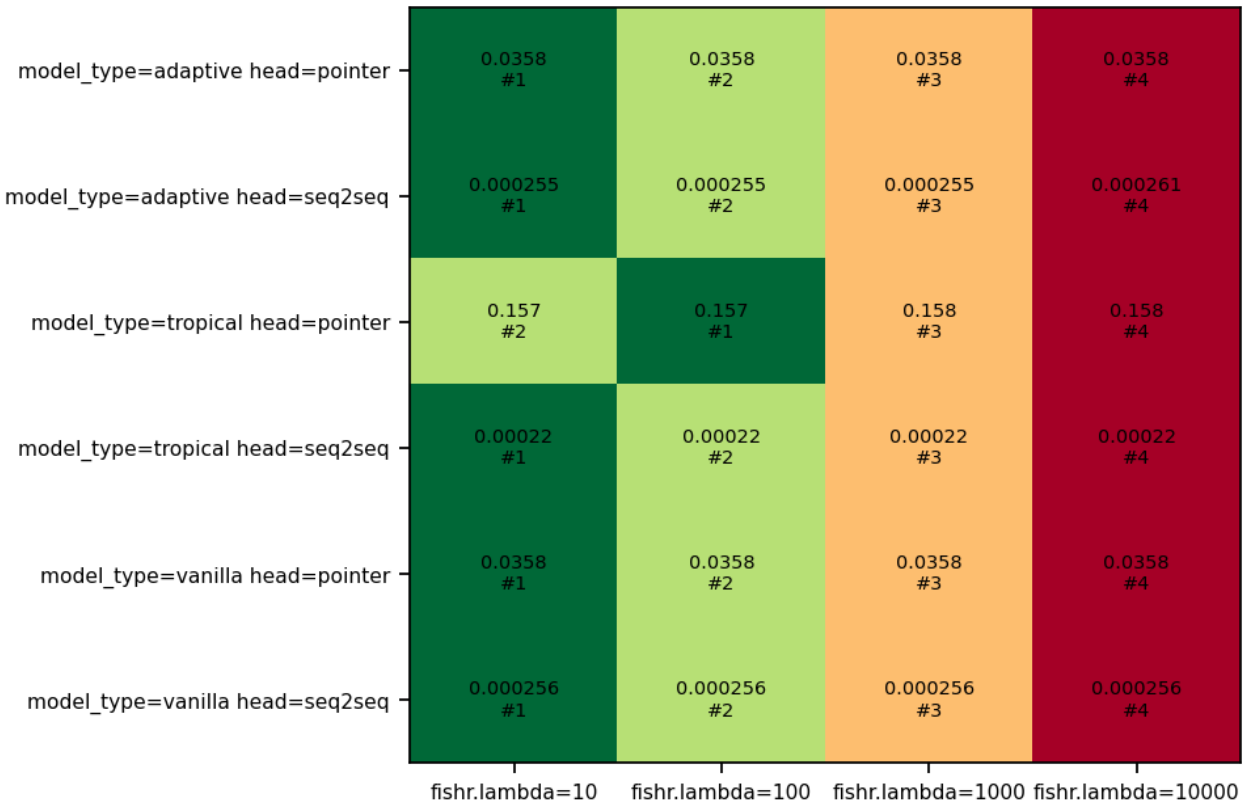


Figure 2: Each cell's seed-mean `id_val_loss` per candidate, with its rank within the cell (#1 is best; green is better).

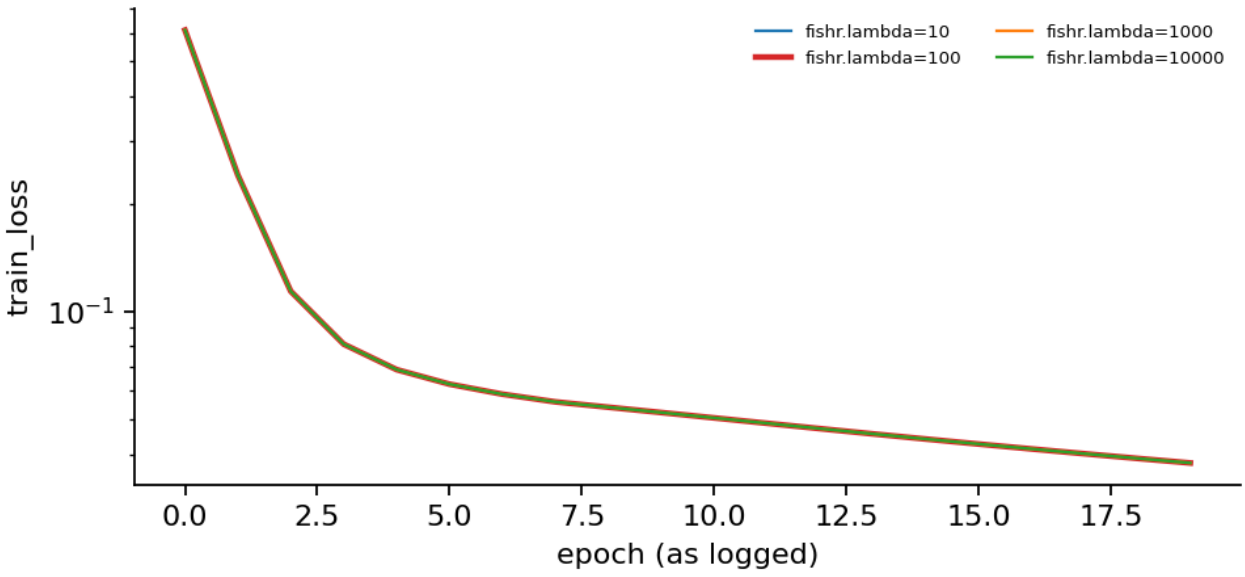


Figure 3: `train_loss` per candidate.

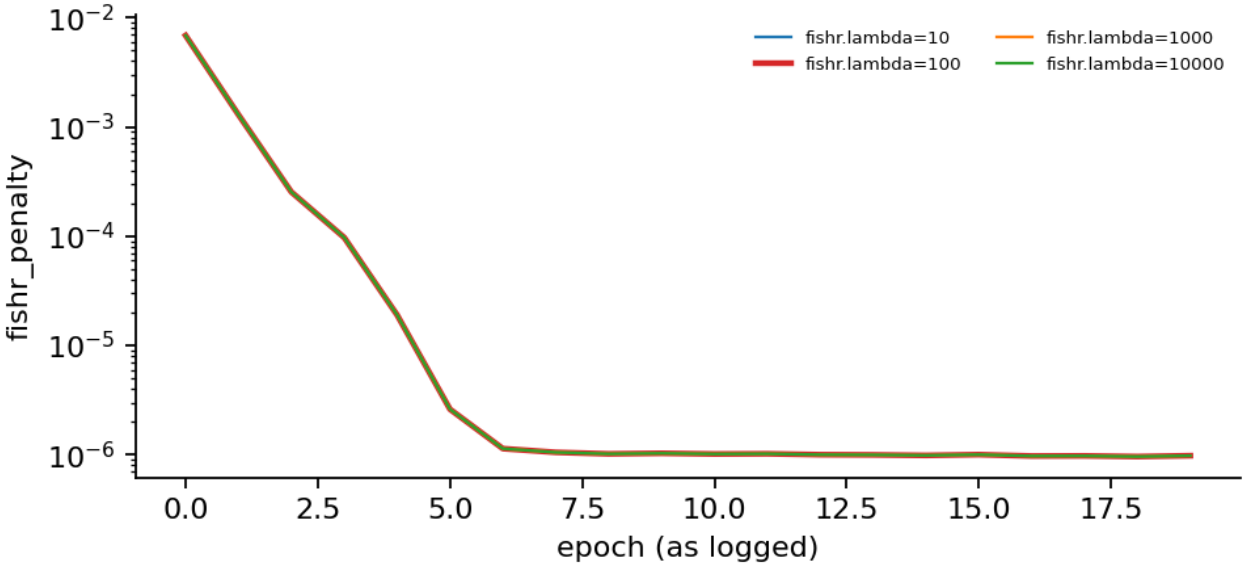


Figure 4: `fishr_penalty` per candidate.

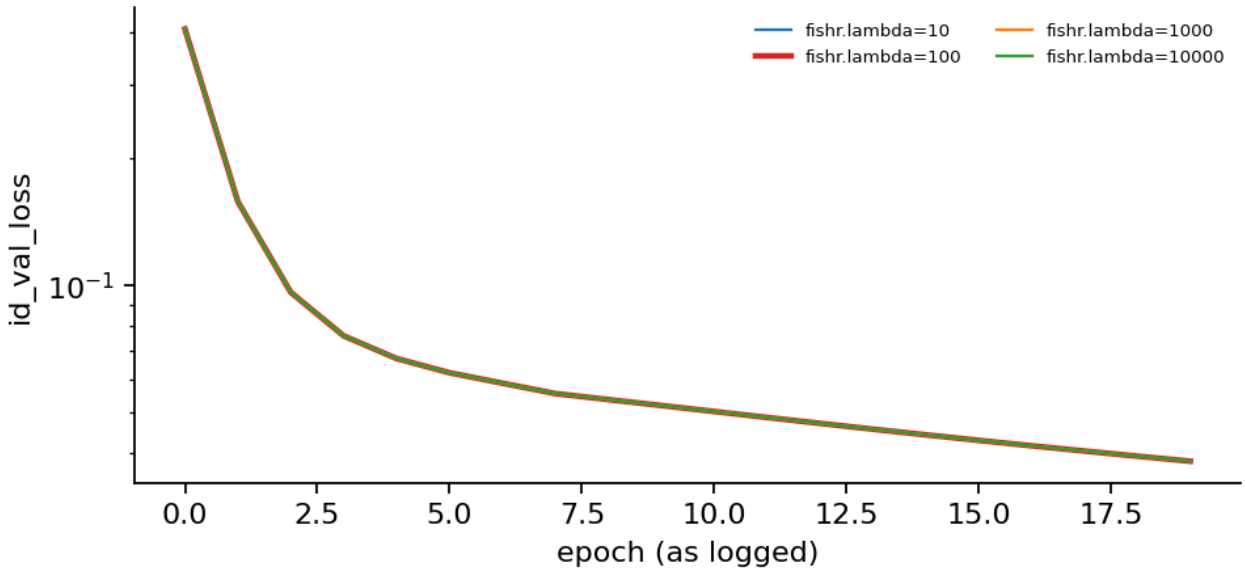


Figure 5: `id_val_loss` per candidate.

7 Provenance

Table 5: Each candidate's runs: base hash, runs, total wall time, commits (+ marks a dirty tree), torch.

candidate	hash	runs	wall (s)	commit	torch
fishr.lambda=10	5f3ea28bbae6	12	3.56e+03	e5ccce8034	2.8.0+cu128
fishr.lambda=100	c7a3d04d3b06	12	3.6e+03	e5ccce8034	2.8.0+cu128
fishr.lambda=1000	84f145eac03b	12	3.6e+03	e5ccce8034	2.8.0+cu128
fishr.lambda=10000	254f24525e3f	12	3.6e+03	e5ccce8034	2.8.0+cu128

Store: `campaigns/tropical/MaxSubsetSum/tune/results`.
Rebuild: `expkit tune-report --config configs/tropical.yaml --profile MaxSubsetSum`
Approve: `expkit approve --config configs/tropical.yaml --profile MaxSubsetSum`

The base the grid would run with.

```
task:
  name: MaxSubsetSumDataset
  value_range:
  - -5
  - 5
  target_range: null
  weight_range: null
  adversarial_range:
  - 10
  - 30
  noise_prob: 0.0
data:
  train_lengths:
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  - 64
  ood_value_range: null
  ood_noise_range: null
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 100
  batch_size: 100
  eval_batch_size: 256
fishr:
  lambda: 100.0
  ema: 0.95
  anneal_iters: 1500
  reset_optimizer: true
```

A Sealed: OOD and test metrics

Read this only after deciding. These were recorded by the tuning runs but never consulted by the selection. Choosing a value because it looks better here would tune on the test distribution.

Table 6: Mean of every OOD or test metric per candidate.

candidate	ood_len_f1	ood_len_exact	ood_val_f1	ood_val_exact	ood_noise_f1	ood_noise_exact	ood_len_f1_L12	ood_len_exact_L12	ood_len_f1_L16	ood_len_exact_L16	ood_len_f1_L64	ood_len_exact_L64
fishr.lambda=10	1	1	0.992	0.988	0.96	0.918	1	1	1	1	1	1
fishr.lambda=100	1	1	0.992	0.988	0.96	0.918	1	1	1	1	1	1
fishr.lambda=1000	1	1	0.992	0.988	0.96	0.918	1	1	1	1	1	1
fishr.lambda=10000	1	1	0.992	0.988	0.96	0.918	1	1	1	1	1	1